

Practical Insights into Data Analysis and Machine Learning

PROGRAM - 2

Develop a program to Load a dataset with at least two numerical columns (e.g., Iris, Titanic). Plot a scatter plot of two variables and calculate their Pearson correlation coefficient. Write a program to compute the covariance and correlation matrix for a dataset. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations.

Objective

To analyze the relationships between numerical variables in a dataset using statistical and visualization techniques.

2. Introduction

In the field of data analysis and machine learning, understanding the relationships between variables in a dataset is a critical step. This program is designed to perform exploratory data analysis (EDA) on a dataset with at least two numerical columns. By leveraging statistical techniques and visualization tools, the program aims to uncover patterns, relationships, and insights hidden within the data.

This program is particularly useful people who want to gain insights into their data before applying more advanced techniques like machine learning or predictive modeling. By combining statistical analysis with visualizations, the program provides a comprehensive understanding of the dataset's structure and relationships, enabling informed decision-making and hypothesis generation. The program focuses on the following key tasks - loading a dataset, visualizing relationships, calculating correlation, covariance and correlation matrices, visualizing correlations.

2.1 Statistical Concepts

Correlation Analysis

- Correlation analysis is a powerful tool for exploring relationships between variables in a dataset. It reveals patterns, strengths, and directions of associations, providing insights that drive further analysis and informed decision-making.
- If one variable increase while the other also increases, it indicates a **Positive Correlation**. For example, there's likely a positive correlation between the number of hours studied and exam scores.
- If one variable increase while the other decreases, it signifies a **Negative Correlation**. there might be a negative correlation between the amount of time spent watching TV and physical activity levels.
- Changes in one variable don't seem to be associated with changes in the other is called **No Correlation**. For example, there might be little to no correlation between shoe size and IQ.

Pearson's correlation

- There are various correlation coefficients, but the most widely used is **Pearson's correlation** (also known as **Pearson's R**). When correlation is mentioned without specifying the type, it typically refers to Pearson's R. However, it's important to note that Pearson's correlation applies only to *numerical data*—detecting relationships in categorical data requires more advanced techniques.
- Correlation measures linear relationships. If the relationship between the variables is non-linear (e.g., curved), the Pearson correlation coefficient might be close to zero even if there's a strong relationship.
- Outliers can significantly affect the correlation coefficient. A single outlier can make a weak correlation appear strong or vice versa.

- Correlation values always fall between -1 and +1:
 - 1 indicates a strong negative correlation.
 - +1 indicates a strong positive correlation.
 - 0 means no correlation, implying no relationship between the variables.

The following table can help interpret correlation values effectively:

Range	Meaning
0.7 to 1.0	Strong positive correlation
0.3 to 0.7	Weak positive correlation
-0.3 to 0.3	Negligible correlation
-0.7 to -0.3	Weak negative correlation
-1.0 to -0.7	Strong negative correlation

Formula for Pearson's Correlation Coefficient

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \times \sqrt{\sum (Y_i - \bar{Y})^2}}$$

Where:

X_i, Y_i are individual data points

$\bar{X}, \bar{Y}$ are the means of X and Y

Correlation Matrix

- A correlation matrix is a table that shows the pairwise correlation coefficients between multiple variables in a dataset. It is a square matrix where each element represents the correlation between two variables. Correlation matrices are widely used in data analysis to understand the relationships between variables.
- The matrix is symmetric because the correlation between variable A and variable B is the same as the correlation between variable B and variable A.

	A	B	C
A	1	0.8	-0.3
B	0.8	1	0.1
C	-0.3	0.1	1

- A and B have a strong positive correlation (0.8).
- A and C have a weak negative correlation (-0.3).
- B and C have almost no correlation (0.1).

Covariance

- Covariance is a statistical measure that describes the extent to which two variables change together. It indicates the direction of the linear relationship between two variables. However, unlike the correlation coefficient, covariance is not standardized, so its value can range from negative infinity to positive infinity, making it difficult to interpret the strength of the relationship.
 - ▶ **Positive Covariance:** Indicates that as one variable increases, the other tends to increase.
 - ▶ **Negative Covariance:** Indicates that as one variable increases, the other tends to decrease.
 - ▶ **Zero Covariance:** Indicates no linear relationship between the variables.

Formula for Covariance

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

- X_i and Y_i : Individual data points.
- $\bar{X}$ and $\bar{Y}$: Means of X and Y , respectively.
- n : Number of data points.

Example: Consider the following dataset

X	Y
1	2
2	3
3	4
4	5
5	6

Calculate the means:

- $\bar{X} = \frac{1+2+3+4+5}{5} = 3$
- $\bar{Y} = \frac{2+3+4+5+6}{5} = 4$

$$\text{Cov}(X, Y) = \frac{(1-3)(2-4) + (2-3)(3-4) + (3-3)(4-4) + (4-3)(5-4) + (5-3)(6-4)}{5}$$

$$\text{Cov}(X, Y) = \frac{(-2)(-2) + (-1)(-1) + (0)(0) + (1)(1) + (2)(2)}{5}$$

$$\text{Cov}(X, Y) = \frac{4 + 1 + 0 + 1 + 4}{5} = \frac{10}{5} = 2$$

The covariance is 2, indicating a **positive** relationship between X and Y.

Differences Between Covariance and Correlation

Aspect	Covariance	Correlation
Range	$-\infty$ to $+\infty$	-1 to +1
Scale	Depends on the units of the variables	Standardized (unitless)
Interpretation	Direction of relationship only	Direction and strength in a standardized way.
Use Case	Less common in practice	Widely used for analysis

1.2 Data Visualization

1.2.1 Scatter plot

- A scatter plot is a type of data visualization that displays the relationship between two numerical variables.
- Each point on the plot represents an observation in the dataset, with the x-axis representing one variable and the y-axis representing the other. Scatter plots are useful for identifying patterns, trends, and correlations between the two variables.
- A scatter plot helps determine whether there is a relationship (positive, negative, or no correlation) between the two variables. For example, if the points tend to rise from left to right, it suggests a positive correlation. If they fall from left to right, it suggests a negative correlation. If there is no clear pattern, it suggests no correlation.
- Scatter plots can reveal outliers, which are data points that deviate significantly from the overall pattern.
- A scatter plot can help determine whether the relationship between the two variables is linear or nonlinear.

1.2.2 Heatmap

- A heatmap is a graphical representation of data where individual values are represented using colors. It is particularly useful for visualizing large matrices, making it easy to identify patterns, relationships, and trends.
- A correlation matrix contains correlation coefficients between multiple variables. Using a heatmap, we can visually interpret these correlations:
 - Strong positive correlation (close to +1) → Darker warm colors (e.g., red/orange).
 - Strong negative correlation (close to -1) → Darker cool colors (e.g., blue).
 - No correlation (around 0) → Neutral colors (e.g., white/light yellow).

This helps quickly identify which variables are strongly or weakly correlated.

- Interpreting a Correlation Heatmap
 - ▶ Diagonal Elements: Always 1 (a variable is perfectly correlated with itself).
 - ▶ Off-Diagonal Elements: Show the correlation between different variables.
 - ▶ Dark red → Strong positive correlation.
 - ▶ Dark blue → Strong negative correlation.
 - ▶ Light colors → Weak or no correlation.
 - ▶ Symmetry: The matrix is symmetric ($r_{XY} = r_{YX}$)

1.3 Program

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the Iris dataset
data = sns.load_dataset('iris')

# Select two numerical columns for scatter plot and correlation calculation
x_col = 'sepal_length'
y_col = 'petal_length'
```

```
correlation = data[['sepal_length', 'petal_length']].corr('pearson')
print("Pearson Correlation Coefficient:\n", correlation)
```

Pearson Correlation Coefficient:

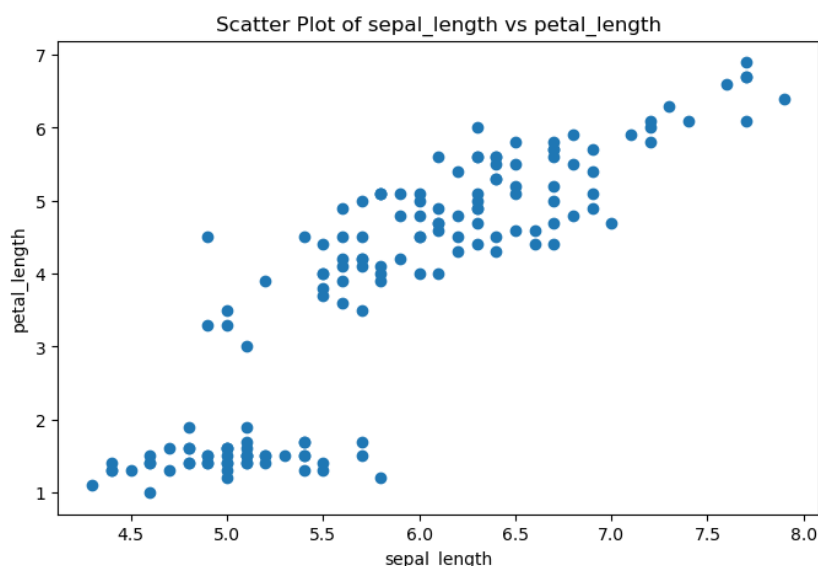
	sepal_length	petal_length
sepal_length	1.000000	0.871754
petal_length	0.871754	1.000000

```
# Compute covariance matrix
covariance = data[['sepal_length', 'petal_length']].cov()
print("Covariance Matrix:\n", covariance)
```

Covariance Matrix:

	sepal_length	petal_length
sepal_length	0.685694	1.274315
petal_length	1.274315	3.116278

```
# Create scatter plot
plt.figure(figsize=(8, 5))
plt.scatter(data[x_col], data[y_col])
plt.xlabel(x_col)
plt.ylabel(y_col)
plt.title(f"Scatter Plot of {x_col} vs {y_col}")
plt.show()
```



```
# Calculate covariance and correlation matrix for a dataset
data_co = data.iloc[:, :-1]
covariance_matrix = data_co.cov()
correlation_matrix = data_co.corr()
print("Covariance Matrix:\n", covariance_matrix)
print("\n Correlation Matrix:\n", correlation_matrix)
```

Covariance Matrix:

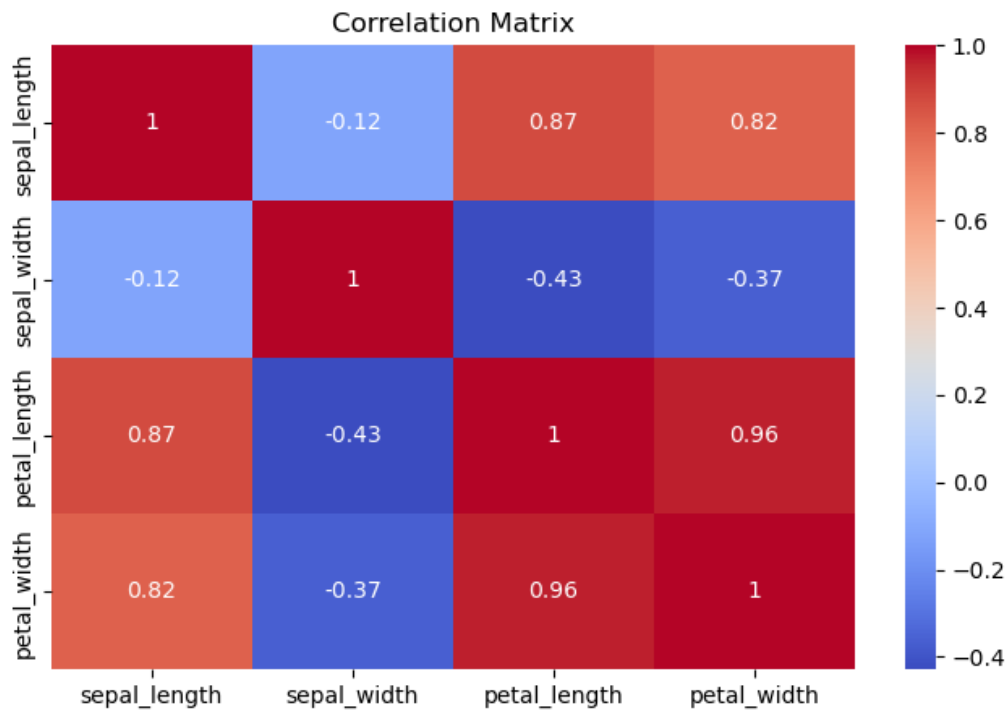
	sepal_length	sepal_width	petal_length	petal_width
sepal_length	0.685694	-0.042434	1.274315	0.516271
sepal_width	-0.042434	0.189979	-0.329656	-0.121639
petal_length	1.274315	-0.329656	3.116278	1.295609
petal_width	0.516271	-0.121639	1.295609	0.581006

Correlation Matrix:

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.117570	0.871754	0.817941
sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

Heatmap: Visualizing the Correlation Matrix

```
# Visualize correlation matrix using heatmap
plt.figure(figsize=(8, 5))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```



Viva Questions

Statistical Concepts:

- What is correlation analysis, and why is it important?
- What is the difference between positive, negative, and no correlation?
- What is Pearson's correlation coefficient, and what does it measure?
- How does Pearson's correlation handle non-linear relationships?
- What is a correlation matrix, and how is it useful?
- What is covariance, and how is it different from correlation?
- What are the key differences between covariance and correlation?

Data Visualization:

- What is a scatter plot, and what information does it provide?
- How can you interpret a scatter plot to determine the relationship between two variables?
- What is a heatmap, and how is it used in data analysis?
- How do you interpret a correlation heatmap?

Advanced Questions:

- How would you handle outliers when calculating correlation coefficients?
- What are some alternatives to Pearson's correlation for non-linear relationships?
- How would you interpret a correlation coefficient of 0.5?
- What are the limitations of using a heatmap for visualizing correlations?
- How would you determine if a correlation is statistically significant?
- What is the difference between a correlation matrix and a covariance matrix?
- How would you use the insights from a correlation matrix in a machine learning model?